JnanaSetu (■■■■■■■■■■) – Sanskrit Knowledge Assis

Technical Report – 2026-11-06

1. Project Overview

JnanaSetu is a locally■run Retrieval■Augmented Generation (RAG) system that answers questions over a curated corpus

2. System Architecture

The architecture consists of four main components:

• Sanskrit NLP pre■processor – script detection, normalization, and transliteration.

• Multilingual Sentence■Transformer embeddings (CPU■only) that embed both queries and document chunks.

• FAISS Index (Inner■Product) storing 94 vectors for the corpus plus uploaded documents.

• Local LLM (GGUF format) loaded via llama.cpp with n_gpu_layers=0.

The retrieval pipeline first checks the similarity score against a confidence threshold; if below, a grounded refusal response

3. Data & Pre■processing

The corpus contains five stories provided as PDF and TXT files in `data/corpus/`. Each document is loaded by `SanskritDoc

4. Embeddings & Retrieval

Embeddings use the `paraphrase■multilingual■MiniLM■L12■v2` model from Sentence■Transformers, configured for CPU

5. CPU■Only LLM Inference

The answer generation stage uses a 0.5B parameter GGUF model (`qwen2.5-0.5b■instruct-q4_k_m.gguf`) loaded through

6. Evaluation Metrics (Actual Results)

Mean Reciprocal Rank (MRR): 0.4394
Precision@3: 0.3939
Recall@3: 0.6364
Answer Faithfulness: 0.2485
Answer Relevance: 0.5333
Avg Retrieval Latency: 60.94 ms
Avg Generation Time: 9.38 s
Avg Tokens / Second: 6.8 tps
Avg Embedding Time: 20.24 ms
Avg CPU Utilization: 96.6%
Avg RAM Consumption: 1052.3 MB

7. Limitations & Future Work

* Retrieval confidence still depends on the similarity threshold; some out■of■domain queries may retrieve the generic `Rag
* The GGUF model is small (0.5■B) and produces slower response times (~9■s). Larger CPU■efficient models could impro
* The current corpus covers only five stories; scaling to larger Sanskrit corpora will require re■indexing and possibly dimens